

AMEER SALMAN M

Forward Deployed Engineer — Data & AI Platforms

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SUMMARY

Data engineer with **8+ years** designing, building and optimising batch and real-time pipelines, lakehouse platforms and automated ETL workflows at petabyte scale. Currently a Forward Deployed Engineer at Databricks, delivering **\$180K+/year in verified cloud savings**, contributing an algebraic migration-validation engine into the open-source **Lakebridge** tool, and building runtime governance for AI agents. Previously seven years at Amazon Pay, cutting reporting SLAs **99% (48 hours to 15 minutes)**, processing **400M+ transactions monthly**, and leading the RBI-mandated regulatory migration of the payments data estate. Deep expertise across **Apache Spark internals, Unity Catalog, Delta Lake and the AWS big-data ecosystem**, combined with customer-facing field engineering.

TECHNICAL SKILLS

LANGUAGES	Python, SQL, Scala, Java, Bash	PLATFORMS	Databricks (Unity Catalog, Delta Lake, Apps, Asset Bundles, System Tables), AWS, Azure
DISTRIBUTED	Apache Spark (internals, Spark SQL, PySpark), EMR, Photon	AWS	Glue, S3, Athena, Redshift, Lambda, MWA, Kinesis, DynamoDB, EC2, CloudWatch, IAM, QuickSight
DATA ENGINEERING	ETL/ELT, real-time & batch streaming, warehousing, dimensional modelling, reconciliation, schema evolution, partitioning	MIGRATION	Lakehouse architecture, JDBC pushdown, algebraic data fingerprinting, Redshift / Teradata / Oracle / Snowflake
FINOPS	DBU cost profiling, execution-plan optimisation, Photon tuning, partition pruning	AI GOVERNANCE	Runtime policy enforcement, OAuth / OBO token brokering, SQL AST parsing, tamper-evident audit, EU AI Act
DEVOPS / IAC	AWS CDK, Terraform, Docker, Git, CI/CD pipelines, automated packaging	TESTING	Pytest, ScalaTest, Testcontainers, GitHub Actions, Maven multi-profile builds

EXPERIENCE

Forward Deployed Engineer · Databricks	JUL 2024 – PRESENT
REMOTE & CUSTOMER ONSITE DATA & AI PLATFORMS	
Workload cost optimisation: profiled and re-engineered high-DBU production Spark and Delta Lake workloads for enterprise accounts, generating \$180,000+/year in verified savings including a 50% compute reduction on a single critical workload, via partition pruning, shuffle minimisation and join restructuring. Algebraic migration validation: engineered a data-fingerprinting framework inside Databricks' open-source Lakebridge migration tool, eliminating the need to move raw tables across the network to prove post-migration parity. Pushdown computation: implemented JDBC pushdown aggregation (sums, power sums, dual-channel collision hashes) on source databases, cutting egress ~48,000x (gigabytes to ~50 KB) and accelerating validation 8.1x on multi-million to billion-row datasets. Exact-difference solver: authored a recovery solver with zero false positives/negatives across 15.8B+ rows, enabling full verification across Redshift, Teradata, Oracle and Snowflake; fixed 3 serialisation and timestamp-drift bugs safeguarding fidelity across all field migrations. Production-scale optimisation sandbox: built a simulation engine on Delta shallow clones and Unity Catalog lineage to test Photon, runtime upgrades and query rewrites against production-volume data without risking live pipelines or duplicating storage. Automated parity verification: implemented bidirectional EXCEPT ALL multiset checking proving 100% byte-level parity across 4.45B+ records while demonstrating 10.19x end-to-end workload acceleration; fail-closed identity boundaries and deferred DBU cost attribution across 30+ pipeline families.	

- **Agentic AI runtime gateway:** architected a multi-workspace security gateway as a Databricks App intercepting, risk-scoring, auditing and remediating autonomous AI-agent operations before production execution, via a **6-checkpoint pipeline** at **sub-30ms** per-checkpoint latency.
- **Zero-trust identity & forensics:** engineered OAuth on-behalf-of token brokering with zero credential sharing plus a hash-chained append-only Unity Catalog audit ledger, delivering compliance evidence for **EU AI Act Article 14** and SOC 2; automated reversal of unauthorised mutations through Delta time travel.
- **Spark connector modernisation:** reverse-engineered Microsoft's archived Spark SQL Server connector for **Databricks Runtime 14.3–18.x** (Spark 3.5 & 4.0), unblocking enterprise Unity Catalog migrations while preserving **~15× bulk write throughput** over generic JDBC; productised with a public-API compatibility layer and automated regression gates, adopted across multiple engagements.

Data Engineer · Amazon Pay / Amazon Payments

OCT 2017 — JUL 2024

BENGALURU & CHENNAI | DATA ENGINEER II ON EXIT

- **Near real-time pipeline architecture:** designed and deployed an end-to-end event-driven pipeline on **S3, Lambda, Glue and Athena**, cutting business reporting SLAs **99% — from 48 hours to 15 minutes** for critical payment data streams.
- **Large-scale ingestion & reconciliation:** engineered multi-source pipelines integrating **15+ upstream services** across **120+ ETL jobs**, processing **400M+ transactions/month** with real-time financial categorisation and reconciliation.
- **Big-data optimisation:** automated high-throughput ETL across multi-terabyte scale on **EMR (Spark), EC2 and S3**, reducing manual operational workload **60% monthly**.
- **Orchestration & elastic compute:** implemented enterprise workflow orchestration on **AWS MWAA** and **AWS CDK**, dynamically provisioning and tearing down transient **EMR clusters** for cost-efficient ETL execution.
- **Infrastructure as Code & CI/CD:** spearheaded IaC adoption using **AWS CDK** across multiple microservices and pipelines; built an internal CI/CD framework for **AWS Glue** packaging dependencies as egg/zip binaries with **Pytest** validation and staged artifact publishing.
- **RBI regulatory migration:** led end-to-end migration of **Redshift clusters, QuickSight dashboards and S3 data lakes** to the **AWS Mumbai region** for Reserve Bank of India payment aggregator and gateway mandates — multi-terabyte cutovers with zero downtime and verified reconciliation.
- **Audit remediation:** conducted deep-dive root-cause analysis on reconciliation discrepancies with banking partners, resolving audit exceptions valued at **₹20+ billion** across India regulatory accounts.
- **Monitoring & alerting:** configured custom **CloudWatch** metrics, automated log parsers and multi-tier alarms for real-time discrepancy detection — zero data loss at **99.9% pipeline uptime**.
- **Operational automation:** built ingestion and validation tooling in **Python and Selenium** for high-volume operational workflows, saving **15+ FTEs** of manual effort monthly; authored SOPs and design documents enabling partner-team self-service.

SELECTED FRAMEWORKS & CONTRIBUTIONS

- **DataPrint (Databricks Lakebridge)** — algebraic migration verifier. 8.1× faster sign-offs, ~48,000× less network egress, validated to 15.8B rows with zero false positives.
- **ShadowClone** — production-risk-free performance sandbox. Enabled confident Photon and runtime upgrades across 30+ pipeline families, demonstrating up to 10.19× execution gains.
- **AgentGuard / LakeGuard** — autonomous AI security gateway. Multi-workspace runtime enforcement with SQL AST firewall, token broker and automated Delta rollback; generates EU AI Act and SOC 2 audit trails.
- **Spark MSSQL Connector** — maintained field distribution. Public-API compatibility layer across Scala 2.12/2.13 and Spark 3.5/4.0, preserving 15× bulk ingestion performance.

EDUCATION

DEGREE

B.E. Mechanical Engineering — Park College of Engineering & Technology, Coimbatore, 2017